Influence of Temporal Representativeness of Satellite Aerosol Products on Direct Aerosol Radiative Effects

Xiaoyi Yang, Ming Zhang, Lunche Wang, and Wenmin Qin

***Abstract*— Direct aerosol radiative effect (DARE) is crucial for atmospheric radiation balance and climate. DARE is usually calculated using instantaneous aerosol optical depth (AOD) from satellite overpasses instead of daily mean AOD. Most satellites pass over twice a day at fixed local times, and these high-frequency data may not be an effective substitute for daily averages, thus affecting calculations of the radiative effects of aerosols. This study undertakes a comprehensive global-scale exploration into the disparities between utilizing satellite AOD products as substitutes for time-averaged values in radiative effect simulations. DARE calculated by instantaneous AOD from satellites was compared with the daily DARE from Aerosol Robotic Network (AERONET). The DARE calculated using MISR AOD showed the lowest root mean square error (RMSE) of 10.3719 $W{\cdot}m^{-2}$. MODIS (Terra/Aqua) performed well in representing daily means, with Terra (RMSE = 11.1417 $W{\cdot}m^{-2}$) and Aqua (RMSE = 10.9722 $W{\cdot}m^{-2}$) showing similar results. In contrast, using VIIRS instantaneous AOD resulted in a higher RMSE of 14.3992 $W{\cdot}m^{-2}$. Since aerosols vary greatly during the day, this article evaluates the representativeness error at different moments of time by comparing the DARE calculated using the daily mean AOD and instantaneous AOD from AERONET. The results show that the DARE from 8:00 a.m. to 2:00 p.m. has the smallest deviation (less than 0.6 $W{\cdot}m^{-2}$), which provides better temporal representativeness for the DARE calculation. The differences in the effect of daily changes in seasonal AOD on DARE calculations were small for all seasons from 10:00 a.m. to 13:00 p.m., and more pronounced in the fall.**

***Index Terms*— Direct aerosol radiative forcing, diurnal time representation, multi–satellite observations.**

I. Introduction

ATMOSPHERIC aerosol refers to solid or liquid particles with a diameter of 0.001–100 μm suspended in the atmosphere, which are widely distributed in the atmosphere with strong spatial heterogeneity. Aerosols have a significant impact on global dimming [1], [2], precipitation [3], atmospheric radiation balance [4], [5], [6], and changes in atmospheric visibility [7], [8]. The direct aerosol radiative effect (DARE), signifies the alteration in the solar radiation flux within the Earth-atmosphere system directly attributable to the presence of aerosols. Aerosol radiative effects are influenced by its optical properties such as aerosol optical depth (AOD), single scattering albedo (SSA), asymmetry (ASY) parameter, Ångström exponent (AE), and so on [9], [10].

Traditionally, studies on aerosol-radiation interactions have relied heavily on daily-averaged AOD data, obtained from both ground-based and satellite observations. Continuous data from a variety of space-based sensors have greatly advanced the understanding of global aerosol distribution, evolution, and sources [11]. These data have deepened insights into aerosols' impact on Earth's radiation balance and their effects on cloud dynamics and precipitation. Satellite remote sensing has become an indispensable tool for monitoring aerosol distributions globally, providing spatially extensive and temporally consistent data that is crucial for understanding the climatic impacts of aerosols [12], [13].

However, the question arises as to whether these daily-resolution data can effectively replace daily averages for more accurate assessments of aerosol radiative effects. Most satellites pass over twice a day at fixed local times, typically around morning and afternoon hours. This sampling frequency may not fully represent the diurnal cycle of aerosol concentrations and optical properties. The representativeness of satellite transit data for estimating radiation effects under clear sky conditions is thus a critical aspect. Ground-based observations more accurately characterize changes in the optical properties of atmospheric aerosols [14], [15] and are often regarded as a benchmark for validating other aerosol measurements [16]. Currently, multiband solar photometry is the most widely used and effective method for detecting the optical properties of aerosols. The Aerosol Robotic Network (AERONET) stands as one of the most extensively utilized site-based datasets in contemporary research [10], [17], [18], [19]. Clear skies provide an ideal scenario to study the pure radiative impact of aerosols, free from confounding factors introduced by clouds. Understanding the extent to which satellite data accurately reflect AOD and its associated radiative effects is critical to atmospheric research [15].

Moreover, these uncertainties in satellite AOD products may propagate further, potentially introducing errors in trend estimations of temperature, global aerosol radiative forcing [20], and $PM_{2.5}$ concentrations [21]. Numerical simulations by Wang et al. [22] demonstrated that a 20% uncertainty in AOD could lead to a 15.4% variation in aerosol direct radiative forcing. The radiative effects of aerosols constitute a significant factor in climate assessment, influencing Earth's radiation balance and climate change through complex

Received 20 January 2025; revised 27 April 2025; accepted 18 May 2025. Date of publication 4 June 2025; date of current version 25 June 2025. This work was supported by the National Natural Science Foundation of China under Grant 42471024 and Grant 42171386. *(Corresponding author: Ming Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3576650

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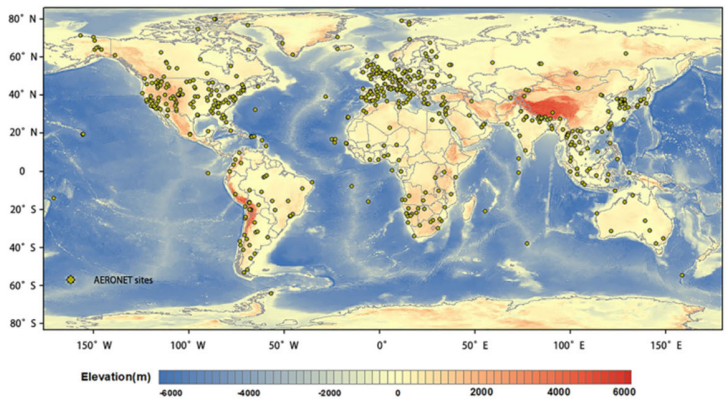


Fig. 1. Geographical locations of the AERONET (yellow dots) used in this work.

and multidimensional mechanisms, primarily manifested in direct radiative effects (DREs) [23] and indirect climate effects [24], [25]. Yu et al. [26] conducted a comprehensive evaluation of aerosol DREs (ADREs) and their climate forcing (DCF) by integrating multisource satellite observations (including MODIS, MISR, and CERES), ground-based monitoring networks (AERONET), and global aerosol models (such as GOCART and SPRINTARS). Quaas et al. [27]. developed a new methodology using data from CERES and MODIS to establish relationships among planetary albedo, cloud properties, and aerosol concentration, and estimate anthropogenic radiative forcing effects, indicating that current global climate models may overestimate the cloud albedo effect. Rap et al. [28] employed a global aerosol microphysics model along with a radiative transfer model to estimate the radiative effects of five natural aerosol sources. Results showed large direct and cloud albedo effects, nonlinear cloud albedo responses, and distinct radiative efficiencies across sources. Luo et al. [29] studied surface solar radiation (SSR) in Nanjing from March 2016 to June 2017 using pyranometer and pyrheliometer observations. SSR's seasonal and diurnal variations were analyzed, and its correlations with AQI, particulate matter (PM_{10}, $PM_{2.5}$), and visibility under different air pollution levels were investigated.

Several studies have quantified uncertainties in radiative effect calculations. McComiskey and Feingold [30] investigated errors in simulated radiative effects caused by aerosol-cloud interactions. Kahnert et al. [31] suggested that using spherical particle models in radiative transfer simulations may constitute one of the primary error sources when quantifying the climate-forcing effects of mineral aerosols. Blaga et al. [32] examined the accuracy of parameterized clear-sky solar irradiance models. Addressing uncertainties in aerosol climate forcing requires urgent research attention.

II. Data and Method

A. Data

The AERONET is a solar-powered weather-hardy network of Sun/sky radiometers that operates over 300 sites worldwide [11]. AERONET provides the angular distribution of sky radiances and aerosol optical properties for a wide range of wavelengths, such as AOD, SSA, and complex refractive index. The AERONET offers a standardization for a ground-based aerosol monitoring and characterization network, with the uncertainty of AOD being about 0.01–0.02 [33]. AERONET Level 2.0 AOD observations have been widely used to validate and correct biases in satellite-derived AOD because of their high data quality, consistency in data processing, broad global coverage, and free data access [34], [35], [36]. This study uses a cloud-screened and quality-assured Level 2.0 inversion product shown in Fig. 1 [37].

Satellites have a stronger global perspective due to their global and repeatable coverage. The more commonly used satellites, such as MODIS, VIIRS, and MISR, have been widely used in atmospheric studies at large scales. Table I summarizes the aerosol product information characteristics used for several of the sensors used in this article, as well as the sensor's emission date and pixel size.

B. Aerosol DRE

SSR plays a crucial role in the global energy cycle and significantly influences the interaction between the Earth's surface and the atmosphere, impacting climate patterns, weather processes, and ecological systems [38]. In this study, we used libRadtran for radiative transfer simulations, which is a software package for radiative transfer calculations [39]. The libRadtran package contains numerous tools that handle various aspects of atmospheric radiative transfer [40]. The radiative transfer solver is based on the sdisort radiative transfer code and accounts for all of the major gaseous constituents in the atmosphere. The input data are shown in Table II, including SSA, ASY, water vapor column, ozone (O_3), and surface albedo were obtained from AERONET. AOD was obtained from AERONET and multisource satellites.

Daily averages of surface aerosol radiative effects are obtained by integrating the instantaneous values of surface aerosol radiative effects over the course of a day in 30-min steps. The DARE for moments when the zenith angle of the celestial sun is greater than 90° is assigned to 0 [41]. The surface aerosol radiation effect is calculated by subtracting the downward shortwave irradiance with aerosols from the downward shortwave irradiance without aerosols. The instantaneous

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TABLE I
SUMMARY OF DATASETS USED IN THIS STUDY

Instrument	Launch Date	Product name	Scientific Data Set	Spatial Resolution
MODIS(Terra)	May2002 10:30 UTC	MOD04_L2	Optical_Depth_Land_And_Ocean (QF = 3)	10 km
MODIS(Aqua)	May2002 13:30 UTC	MYD04_L2	Optical_Depth_Land_And_Ocean (QF = 3)	10 km
VIIRS	October 2011 13:30 UTC	VIIRS_AERDT	Optical_Depth_Land_And_Ocean	6 km
MISR	December 1999 10:30 UTC	MISR_AM1_AS_AEROSOL	Aerosol_Optical_Depth(QF = 3)	4.4 km

TABLE II
RADIATIVE TRANSFER ANALOG INPUT PARAMETERS

Data sources	Input parameter
AERONET	Aerosol optical depth (AOD)、Single scattering albedo (SSA)、asymmetric parameter (ASY)、Water vapor column、Ozone (O_3)、Surface albedo
Satellite	Aerosol optical depth (MODIS AOD/MISR AOD/VIIRS AOD)

aerosol DARE is defined as follows:

$$\mathrm{DARF} = \mathrm{F}^{\downarrow}_{\mathrm{aerosol}} - \mathrm{F}^{\downarrow}_{\mathrm{no-aerosol}} \tag{1}$$

where $\mathrm{F}^{\downarrow}_{\mathrm{aerosol}}$ and $\mathrm{F}^{\downarrow}_{\mathrm{no\text{-}aerosol}}$ are the upward flux in an atmosphere with and without aerosols.

C. Matchup Methodology

This study deals with the evaluation of three AOD products (i.e., VIIRS, MISR, and MODIS). AERONET stations offer AOD products in the 440, 675, 870, and 1020 nm bands, but not in the 550 nm band. To match the central wavelength (550 nm) of the AOD products, the AERONET AOD value at 550 nm was estimated from the AERONET AOD values between 440 and 675 nm using the Angström exponent (α), which can be expressed as follows:

$$\mathrm{AOD}_{550} = \mathrm{AOD}_{\lambda}\left(\frac{550}{\lambda}\right)^{-\alpha} \tag{2}$$

where λ is the wavelength, AOD_{λ} is the AOD at λ. It has been shown that the uncertainty introduced by this interpolation method is negligible.

However, AERONET data often have missing values due to cloud cover, instrument maintenance, and other limitations [42]. This can introduce uncertainty if users do not apply screening criteria when using AERONET data as a reference. In this study, quality control was performed on AERONET data when calculating the mean value over a period of time. Instantaneous matched AOD is defined as AERONET data within ±30 min of the satellite overpass. Daily AERONET AOD data are theoretically the average of all AERONET measurements taken during the day, but data gaps exist. Different studies have proposed varying time windows for different seasons. In this study, the time windows for collecting daily mean AOD from AERONET were 7:00 to 17:00 in spring, 6:30 to 18:00 in summer, 7:30 to 16:30 in autumn, and 8:00 to 16:30 in winter. Midday is defined as 11:00 to 13:00, with morning and afternoon covering the rest of the daily time window. The thresholds were set to three data points in the morning, two data points at noon, and three data points in the afternoon. This control better reflects the daily AOD variation pattern. The spatial matching of the satellites to the AERONET sites was obtained by selecting the satellite AOD values for a 3 × 3-pixel sampling window centered on each AERONET site and calculating the average of the valid AOD data (at least 20% of the pixels were valid) within the window.

This study comprises three main research components: first, we evaluated the impact of satellite retrieval uncertainties on DARE calculations by comparing instantaneous satellite AOD with AERONET station measurements averaged over a ±30-min window centered on the satellite overpass time. Second, we conducted a comparative analysis between instantaneous satellite AOD and daily averaged AERONET AOD to assess the combined effects of temporal representativeness errors and instrumental measurement uncertainties on DARE estimation. Finally, leveraging high-temporal-resolution AERONET observations, we quantified the influence of substituting time-specific AOD measurements for daily mean values in DARE calculations, thereby isolating the specific contribution of temporal representativeness to DARE estimation errors.

D. Evaluation Metrics

Commonly used error measures are used to assess and compare the results of predictive models. In this study, a 1-D linear regression simulation metric was selected to fit the DARE calculated using AERONET AOD and satellite AOD, respectively,

$$\mathrm{DARE}_{\mathrm{satellite}} = a \times \mathrm{DARE}_{\mathrm{AERONET}} + b. \tag{3}$$

The following assessment metrics were used to quantify the accuracy of the DARE: 1) relative mean error (RME) (above and below 1.0 indicate overestimation and underestimation, respectively); 2) correlation coefficient (R), which reflects

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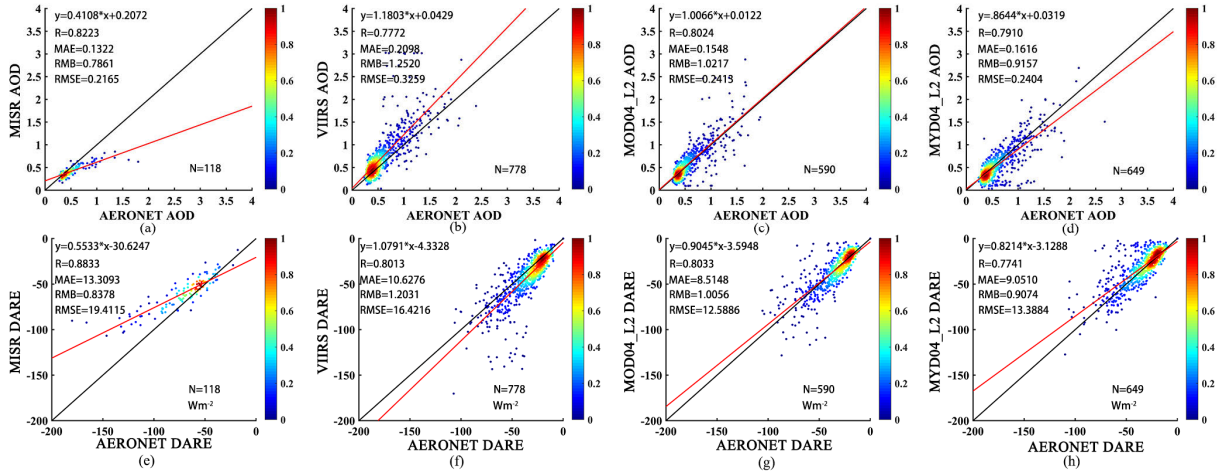


Fig. 2. Density plots between the remote sesing-derived and AERONET-observed AODs at (a) MISR, (b) VIIRS, (c) MOD, and (d) MYD, which were generated using the entire global AERONET data archive between 2015 and 2019 and concurrent satellite AOD datasets. And comparison with scatter plots of clearing radiative effects calculated using different satellite AOD products and AERONET AOD products, (e) MISR, (f) VIIRS, (g) MOD, and (h) MYD.

the deviation between the observed and modeled values; 3) root mean square error (RMSE); and 4) mean absolute error (MAE).

III. Results

A. Clear-Sky Radiation Effects: Evaluating Satellite Transit Consistency Over 1 h

This section will focus on a detailed comparison between the AERONET AOD values that are instantaneously matched with satellite data and the AOD values observed by the satellite. Following this comparison of AOD values between satellite and ground-based measurements, we then transition to examining the radiative effects influenced by these aerosol loads.

Fig. 2 shows the combined statistics of MISR, VIIRS, and MODIS (Terra and Aqua) AOD versus AERONET-based AODs for the entire study region for the period 2015–2019 for both land and ocean. MISR, VIIRS, and MODIS (Terra and Aqua) eligible counterparts are 118, 778, 590, and 649, respectively. The low number of matches can be attributed to the frequent cloud contamination and the relatively limited number of accessible sites. The MISR satellite showed a good fit with the ground-based stations, with an R value of 0.8223, but the relative mean bia (RMB) was 0.7861, indicating a persistent underestimation by the MISR satellite. The VIIRS satellite had an R value of 0.7772, the lowest among the four satellites, with an RMSE of 0.3259. Similar to the MISR, the RMB for VIIRS is significantly overestimated values in the VIIRS satellite data. The MODIS morning (Terra) and afternoon (Aqua) satellites showed similar fitting qualities with R values of 0.8024 and 0.7901, respectively, and RMSE values of 0.2413 and 0.2404. However, the morning star (Terra) tended to overestimate, while the afternoon star (Aqua) tended to underestimate.

Similar to the AOD comparisons, MODIS Terra and Aqua also show quite similar fitting results for radiation effects under clear-sky conditions, with R of 0.8033 and 0.7741, respectively, and RMSE close to each other at 12.5886 and 13.3884 W·m^{-2}. There is a noticeable difference in the RMB, with values of 1.0056 and 0.9074, respectively, indicating that the Terra satellite generally performs better than the Aqua satellite. The distribution of points for the MISR satellite appears less concentrated, despite having the highest R of 0.8833 among the four satellites. However, its RMSE is relatively high at 19.4115 W·m^{-2}, and the RMB is 0.8378, indicating that the overall deviation of the MISR satellite is significant. The VIIRS satellite has R of 0.8013, RMB of 1.2031, and RMSE of 16.4216 W·m^{-2}, which is slightly higher than the MODIS satellites. Overall, its performance is considered good. There are noticeable outliers in the bottom right corner, which correspond to the deviations seen in the AOD comparison.

B. Clear-Sky Radiation Effects: Evaluating the Performance of Satellites in the Representation of Daily Averages by Instantaneous Values

Aerosol concentration and distribution change throughout the day due to factors like weather conditions and human activities. Using satellite instantaneous AOD values instead of daily averages in radiative calculations has unclear implications. This section explores the effects of using satellite instantaneous AOD values instead of daily mean AOD values for calculating clear-sky radiative effects as shown in Fig. 3. The number of valid comparisons for MISR, VIIRS, and MODIS (Terra and Aqua) are 1953, 10 325, 9519, and 8969, respectively.

The MISR satellite exhibits better performance with daily averages compared to instantaneous matches, achieving an R value of 0.8135 and an RMSE of 0.1947. However, the RMB value is 0.8705, indicating a noticeable underestimation, as evident from the plot. In contrast to MISR, the VIIRS satellite shows a degree of overestimation, consistent with the results of the instantaneous matches. VIIRS has the lowest R value among the four satellites at 0.7901, along with an RMSE

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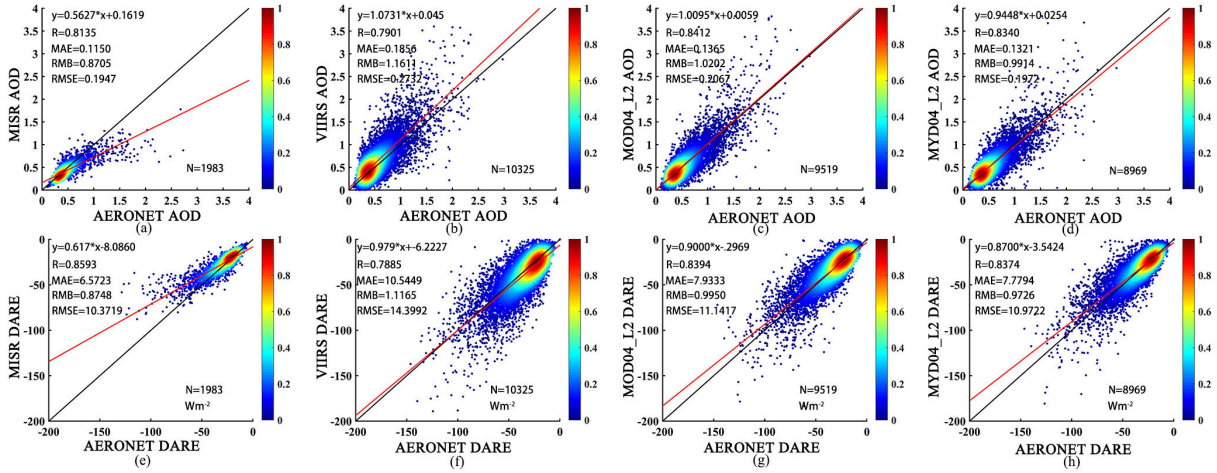


Fig. 3. Density plot between remotely sensed instantaneous AOD and AERONET daily mean AODs at (a) MISR, (b) VIIRS, (c) MOD, and (d) MYD. And comparison with scatter plots of clearing radiative effects calculated using different satellite AOD products and AERONET AOD products, (e) MISR, (f) VIIRS, (g) MOD, and (h) MYD.

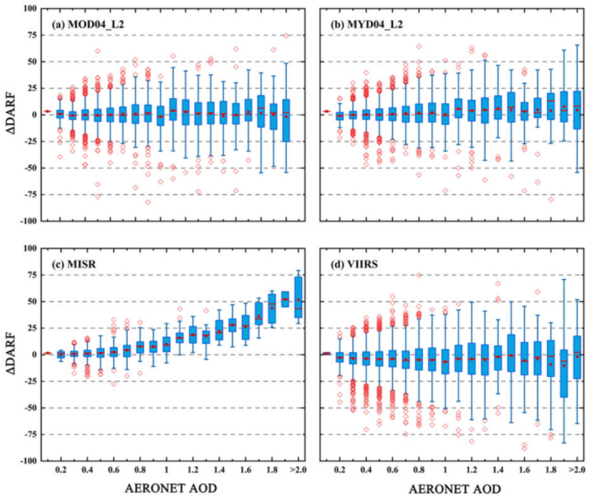


Fig. 4. Validation boxplot of AOD size against DARE. (a) MOD04_L2. (b) MYD04_L2. (c) MISR. (d) VIIRS.

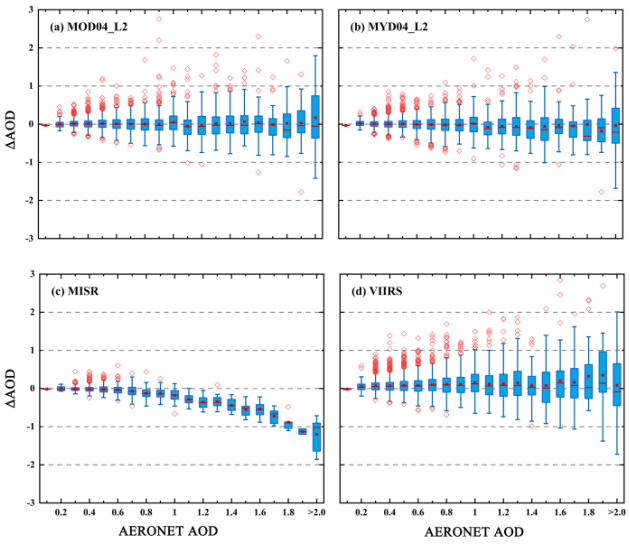


Fig. 5. Validation boxplot of ΔAOD (remote AOD-AERONET aod) size against AOD. (a) MOD04_L2. (b) MYD04_L2. (c) MISR. (d) VIIRS.

of 0.2732. The overall performance of the MODIS morning and afternoon satellites remains similar, with the morning satellite exhibiting slightly higher bias than the afternoon satellite. The R values for the morning and afternoon satellites are 0.8412 and 0.8340, respectively, and their RMSE are 0.2067 and 0.1932.

When using satellite instantaneous values instead of daily averages to calculate the radiative effect under clear-sky conditions, VIIRS performs relatively poorly, with an R value of 0.7885 and an RMSE of 14.3992 W·m^{-2}. The MODIS satellites demonstrate a more balanced distribution, with the morning and afternoon satellites still showing little difference. Compared to AOD, both exhibit an underestimation of the radiative effect. In comparison, the MODIS Terra performed slightly better with an R of 0.8394 and an RMSE of 11.1417 W·m^{-2}. The MODIS Aqua had an R of 0.8374 and an RMSE of 10.9722 W·m^{-2}. The MISR satellite has an R value of 0.8593 and an RMSE of 10.3719 W·m^{-2}.

Fig. 4 illustrates the variance due to different AOD sizes when satellite AOD is used instead of daily mean AOD to calculate the DARE. Specifically, the MODIS Terra and Aqua show good performance at all AOD levels, although the MODIS Aqua shows a slight increase in bias at high AOD values compared to the MODIS Terra. The MISR satellites, on the other hand, show an overall trend: the daily DARE values calculated using satellite AOD instead of daily average AOD gradually increase with increasing AOD values, and perform best at AOD values below 1. In contrast, the error in the daily DARE computed using the VIIRS instantaneous AOD fluctuates more, and this error increases with increasing AOD values.

As shown in Fig. 5, we conducted a comparative analysis between satellite AOD product diurnal representation errors and AERONET AOD measurements, which reveals that MISR AOD products exhibit significant underestimation with increasing AOD values. This finding is consistent with the

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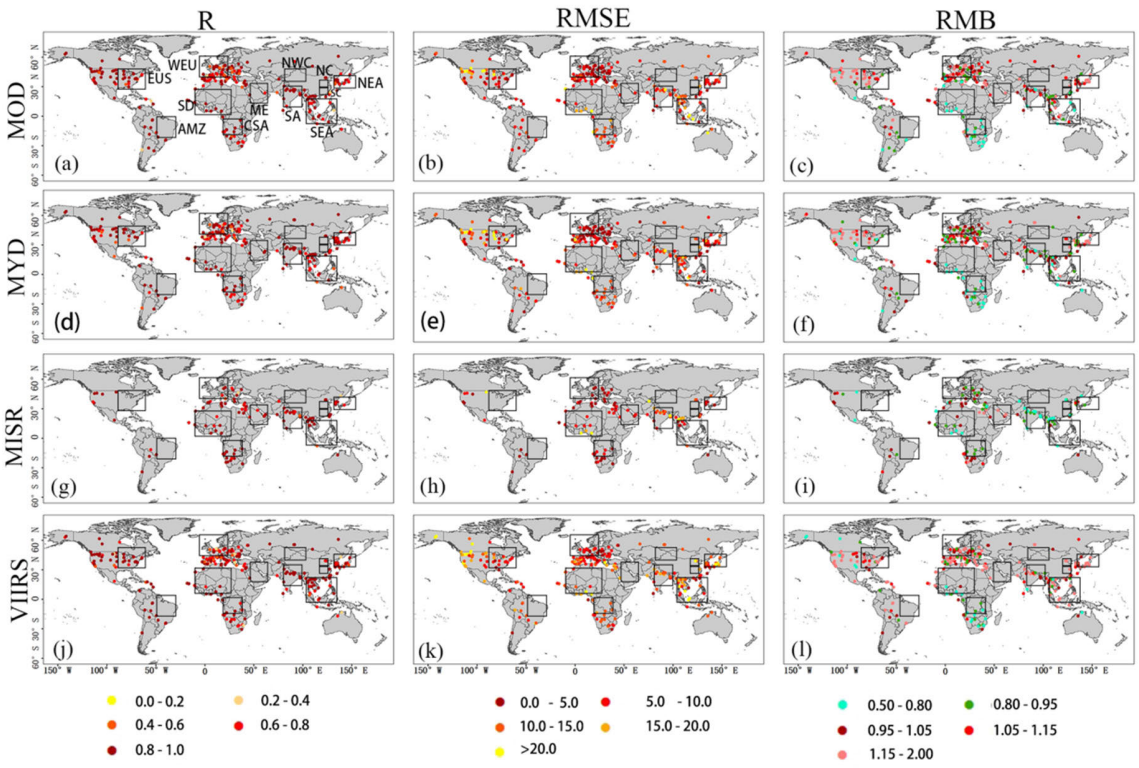


Fig. 6. Comparison of the radiative effects of clearing calculated using the MODIS AOD, the MISR AOD and the VIIRS AOD with the AERONET global AOD at each site, respectively, for the years 2015–2019: site performance maps for (a) MOD correlation coefficient (R), (b) MOD root-mean-square error (RMSE), (c) RMBs; (d)–(f) same as (a)–(c), but for MYD; (g)–(i) same as (a)–(c), but for MISR; and (j)–(l) same as (a)–(c), but for VIIRS. The black rectangles represent the 12 regions of interest selected in this study: northeast Asia (NEA), northern China (NC), southern China (SC), southeast Asia (SEA), northwest China (NWC), southern Asia (SA), middle east (ME), western Europe (WEU), Sahel Desert (SD), central Africa (CF), eastern United States (EUS), and Amazon (AMZ) zone.

TABLE III
DARE STATISTICS CALCULATED USING AERONET AOD DAILY AVERAGES VERSUS SATELLITE AOD FOR EACH OF THE 12 REGIONS OF INTEREST

Geographic Area		SD	EUS	AMZ	WEU	CF	ME	SA	NC	SC	SEA	NEA
MOD04_L2	N	622	145	98	612	533	111	1657	337	10	779	686
	R	0.84	0.93	0.80	0.75	0.79	0.88	0.82	0.85	0.97	0.87	0.81
	RMSE	18.08	10.64	10.12	5.89	13.61	7.99	10.46	10.93	5.70	12.59	8.08
	RMB	0.76	1.20	0.89	1.04	0.75	1.14	1.01	1.10	1.02	1.12	1.09
MYD04_L2	N	587	98	71	623	474	111	1569	333	16	676	661
	R	0.84	0.89	0.85	0.73	0.79	0.86	0.83	0.85	0.96	0.85	0.80
	RMSE	18.77	14.93	9.35	6.07	12.15	8.53	9.95	9.96	3.84	10.93	7.80
	RMB	0.73	1.13	0.86	1.01	0.76	1.04	0.98	1.08	1.07	1.03	1.08
MISR	N	198	28	11	113	111	207	271	52	-	105	90
	R	0.85	0.87	0.96	0.84	0.92	0.77	0.84	0.86	-	0.93	0.84
	RMSE	15.50	15.88	4.59	4.23	6.57	7.81	10.54	14.02	-	8.53	5.33
	RMB	0.79	0.83	0.94	0.87	0.92	1.09	0.83	0.74	-	0.88	0.88
VIIRS	N	699	117	80	678	527	80	1780	366	27	808	730
	R	0.80	0.89	0.85	0.71	0.79	0.91	0.81	0.74	0.97	0.85	0.68
	RMSE	19.08	14.59	10.55	8.69	13.23	17.95	13.56	21.12	5.705	16.10	13.62
	RMB	0.77	1.25	0.95	1.18	0.77	1.35	1.13	1.43	1.11	1.25	1.33

validation results reported by Fan et al. [43], Si et al. [44], Choi et al. [45], and Jiang et al. [46]. Their research demonstrated systematic biases in MISR retrievals: an overestimation tendency at very low AOD conditions contrasted with underestimation at higher AOD ranges. Specifically, Fan's study [43] quantified these biases, showing substantial

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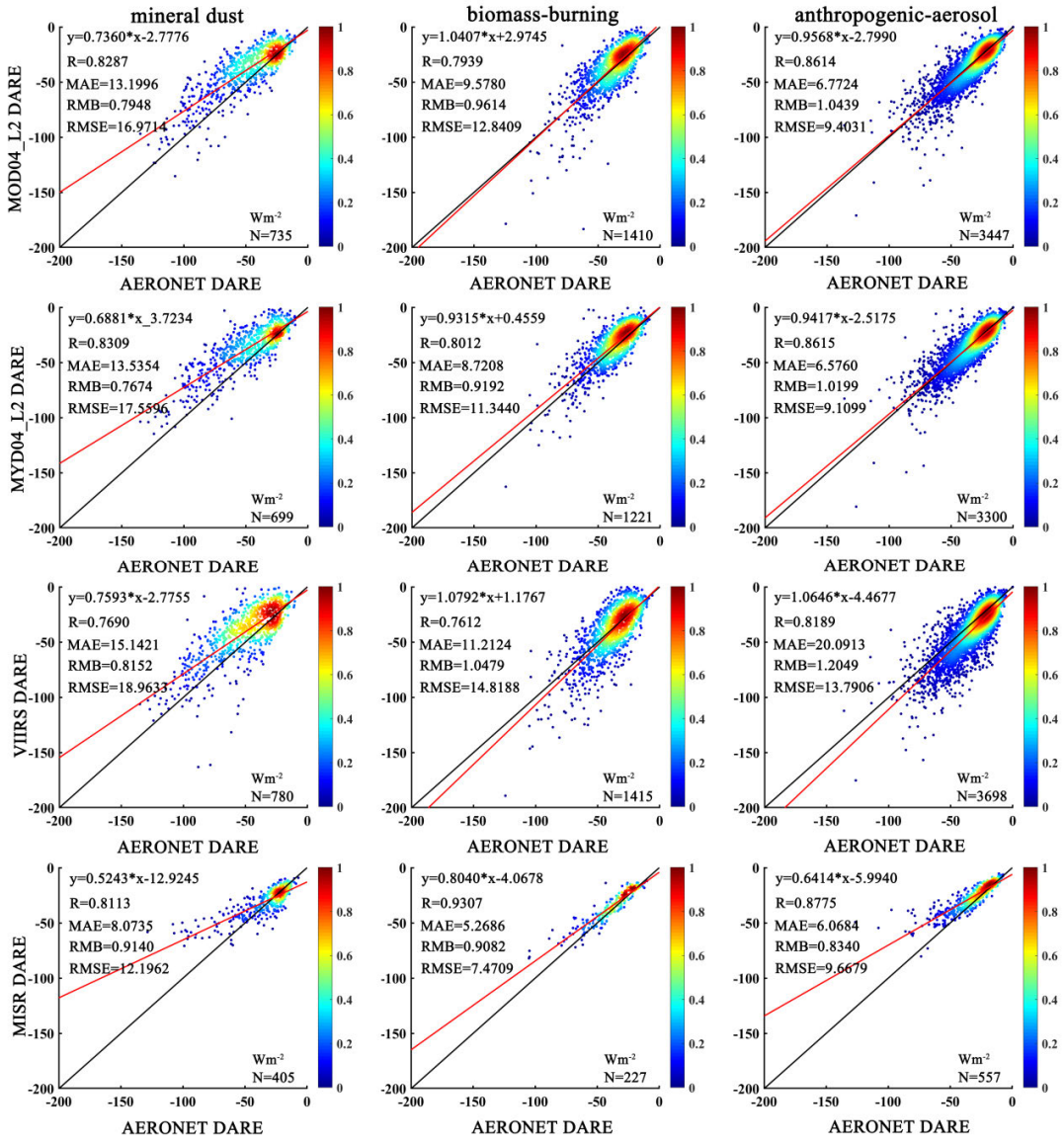


Fig. 7. Influence of the daily temporal representativeness of satellite AOD products on DARE based on different aerosol types.

underestimation of fine-mode particles over continental regions (RMB = 0.588) and marine environments (RMB = 0.734) under high AERONET AOD conditions, while coarse-mode AOD displayed poorer performance with notable overestimation (RMB = 1.835) in low AOD scenarios.

Site-level assessments were used to illustrate the details of the representation of different regions. Fig. 6 shows the representation of sites for each satellite. Fitting of the clear-sky radiative effects of MISR to Earth-based observations: more than 86.8%, 78.8%, 57.6%, and 24.1% of sites had an R larger than 0.6, 0.7, 0.8, and 0.9, respectively; 91.1%, 79.1%, 54.9%, and 17.1% of sites had an RMSE lower than 20, 15, 10, and 5 (Wm^{-2}), respectively; and over 62.2% and 37.7% of sites had an RMB $>$ 1 and RMB $<$ 1, respectively. The difference in R between MODIS Terra and Aqua is not significant, but VIIRS is clearly inferior to MODIS in terms of RMSE, with 10.1% and 10% of MODIS (Terra and Aqua) RMSE $>$ 15 $W{\cdot}m^{-2}$, respectively, compared to 23.5% for VIIRS. The VIIRS satellites have 73.5% and 26.5% RMB $<$ 1 and RMB $>$ 1, respectively, which are overestimated in most areas. Western Europe (WEU) has a low RMSE, but Southeastern Asia (SEA) has a high RMSE. The different satellites are mostly underestimated in the South America and South Africa region. In East Asia, the clear-sky radiative effects calculated by both MODIS and VIIRS show an overestimation, while the clear-sky radiative effects calculated by MISR satellites show an underestimation.

The impact of daily temporal representativeness of satellite AOD products on DARE across 12 regions is shown in Table III. The northwestern China (NWC) region had an extremely limited number of station matches and was therefore excluded from the discussion.

The performance of satellite sensors exhibits pronounced regional variability. In the Amazon (AMZ) region, the MISR sensor demonstrates exceptional proficiency, achieving R of 0.96 and RMSE of 4.59. However, MISR's error metrics

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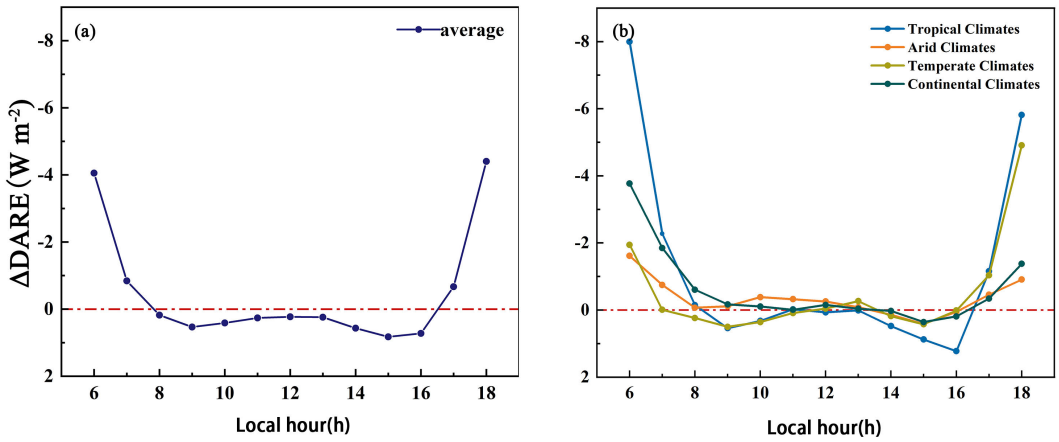


Fig. 8. Clear-sky radiative effects calculated using daily average AOD minus clear-sky radiative effects calculated using hourly AOD of (a) 2015–2019 ave AOD and (b) different KG climate zone AOD.

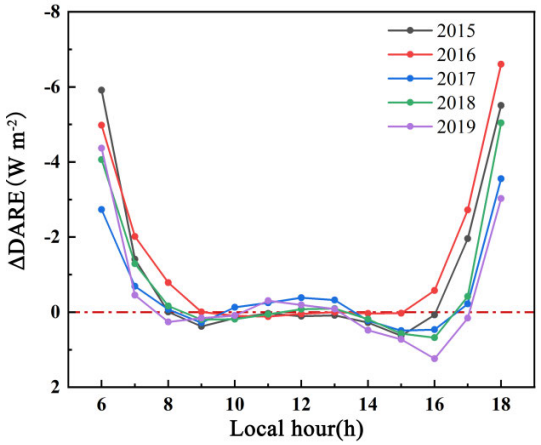


Fig. 9. Clear-sky radiative effects calculated using daily average AOD minus clear-sky radiative effects calculated using hourly AOD of 2015–2019 AOD.

in the Sahara Desert (SD) are notably elevated (RMSE = 15.50 W·m^{-2}) relative to other regions, underscoring challenges in capturing aerosol dynamics over hyper-arid surfaces. Conversely, the VIIRS sensor exhibits superior accuracy in South China (SC) and the Amazon (R = 0.97 and 0.85, respectively), yet reveals extreme discrepancies in the SA and NC, with RMSE values escalating to 13.56–21.12 W·m^{-2} and substantial overestimation (RMB = 1.13–1.43). These anomalies may reflect VIIRS's heightened sensitivity to high-albedo surfaces or calibration biases in specific geographic contexts.

The WEU region manifests a unique error profile: while correlation coefficients (R = 0.71–0.84) are lower than those observed in SC, RMSE values exhibit remarkable variability across satellite platforms (4.23–8.69 W·m^{-2}), with a pronounced elevation in VIIRS data (RMSE = 8.69 W·m^{-2}). This divergence likely arises from the intricate diurnal cycles of aerosols in temperate marine climates, where AOD peaks are modulated by seasonal land-sea breeze circulations and diurnal anthropogenic emission patterns. Such dynamics complicate the representation of daily mean aerosol loading using single-overpass satellite observations, emphasizing the need for temporally adaptive retrieval algorithms in this climatic zone.

These results indicate that the overall statistics for individual satellites are good, but there are some sites that underperform consistently. Overall, the MODIS satellite AOD is well represented in the calculation of clear-sky radiative effects, with a low RMSE and no significant misestimation. The numerical performance of the MISR satellites is very good, but there are underestimations of a large number of stations, which have an impact when it comes to practical applications. VIIRS has the largest RMSE of them all, along with a more pronounced overestimation, and does not perform as well as MODIS and MISR.

The MISR data gaps in North America, caused by other constraints on the calculation of radiation, may account for the larger proportion of MISR underestimates. Despite a high percentage overestimation, VIIRS shows less bias according to the RMB. Previous investigations have highlighted that the spatial variability of aerosol distributions can introduce biases in assessment outcomes, potentially undermining their reliability [47]. Moreover, the precision of AOD values derived from satellite data is frequently challenged by the intricate diversity of terrestrial surfaces, which can complicate the retrieval algorithms and affect the quality of the resulting datasets [48].

Fig. 7 evaluates clear-sky DARE errors arising from diurnal temporal representativeness differences in satellite AOD products across three major aerosol types by comparing DARE derived from satellite instantaneous AOD with that calculated using AERONET daily mean AOD. In dust-dominated regions, all four satellite AOD products led to DARE underestimation due to diurnal representativeness limitations. MOD04_L2 and MYD04_L2 exhibited similar performance with R values ranging from 0.82 to 0.83 and RMSE values between 16.97 and 17.55 W·m^{-2}, while VIIRS showed the largest errors (R = 0.77, RMSE = 18.96 W·m^{-2}) and MISR achieved the smallest RMSE of 12.19 W·m^{-2}. In biomass burning-dominated regions, DARE errors caused by satellite AOD diurnal representativeness were smaller compared to dust regions. MISR delivered the best performance with an R value

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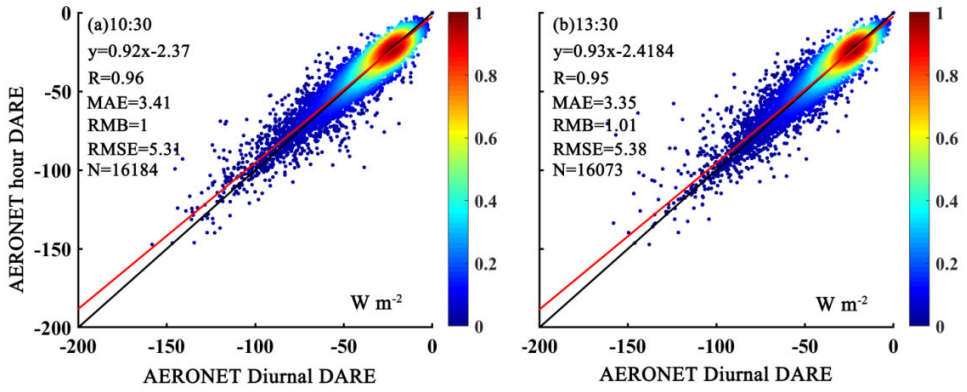


Fig. 10. Comparison of DARE calculated from AERONET AOD hourly means (±30 min around satellite overpass) and daily means.

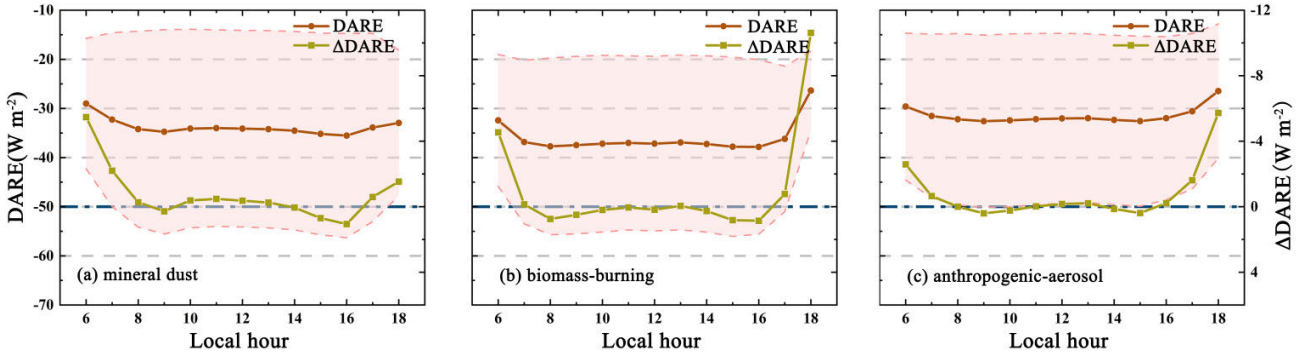


Fig. 11. Differences in clear-sky DARE calculated based on the daily variations of AOD for different aerosol types, red is DARE calculated using different time periods, green is the difference in DARE from that calculated using daily AOD (DARE calculated from the daily average AOD—DARE calculated from the hourly average AOD).

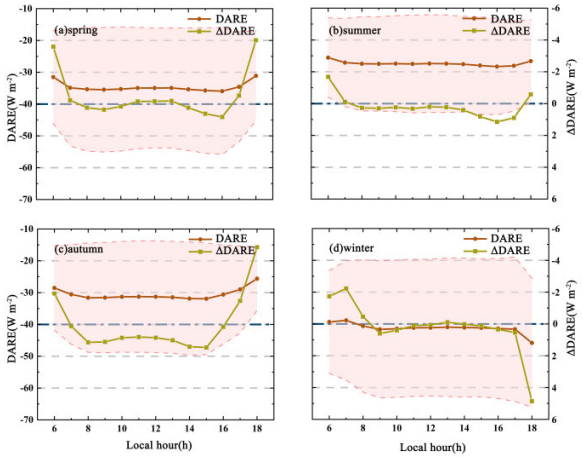


Fig. 12. Impact of daily changes in AOD on DARE calculations.

of 0.93 and RMSE of 7.47 W·m^{-2}, followed by MYD04_L2 (R = 0.80, RMSE = 11.34 W·m^{-2}) and MOD04_L2 (R = 0.79, RMSE = 12.84 W·m^{-2}), while VIIRS still produced the highest errors (R = 0.76, RMSE = 14.81 W·m^{-2}. In anthropogenic aerosol-dominated regions, MOD04_L2, MYD04_L2, and MISR generated relatively small DARE errors with R values from 0.86 to 0.87 and RMSE values between 9.10 and 9.66 W·m^{-2}. However, MOD04_L2 and MYD04_L2 tended to overestimate DARE, whereas MISR underestimated it. VIIRS displayed more pronounced errors (R = 0.82, RMSE = 13.79 W·m^{-2}). Overall, the MISR AOD product achieved the highest DARE accuracy in–biomass-burning regions and the smallest errors in dust-dominated areas. The MODIS AOD products (MOD04_L2/MYD04_L2) performed best in anthropogenic aerosol regions, demonstrated moderate accuracy in biomass-burning regions, and matched MISR's precision in dust regions. In contrast, the VIIRS AOD product exhibited the largest errors across all aerosol types, indicating overall inferior performance.

C. Impact of Daily AOD Variations on Daily DARE

The concentration and characteristics of aerosols are influenced by various factors such as terrain and climate, leading to significant variations within a single day. To investigate the representativeness of AOD in calculating clear-sky radiative effects at different times, this study compares the radiative effects calculated using hourly AOD values with those calculated using daily average AOD values, as shown in Fig. 8. As shown in Fig. 8(a), by averaging the data from 2015 to 2019, it is observed that using AOD values from 8:00 a.m. to 2:00 p.m. to replace the daily average for calculating clear-sky radiative effects results in the smallest differences. In contrast, deviations are larger in the

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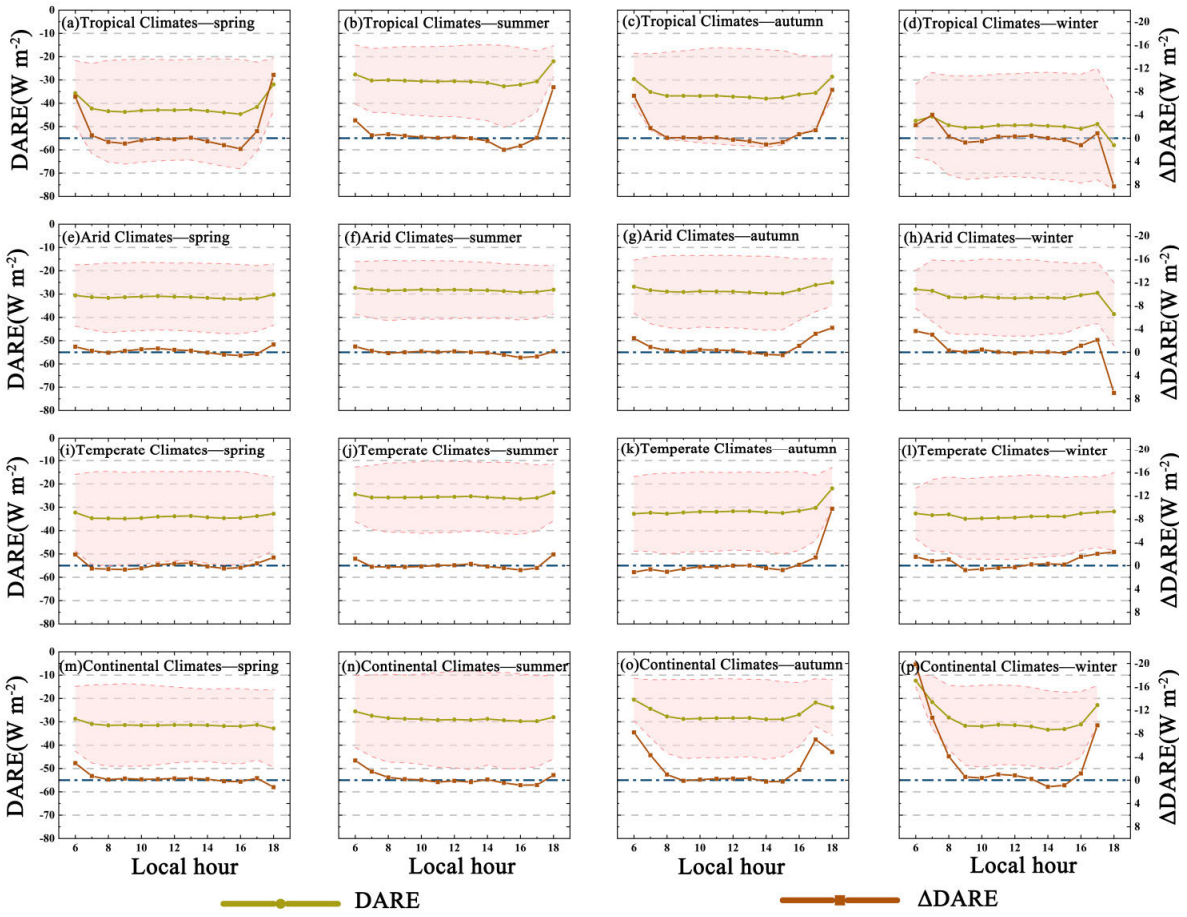


Fig. 13. Impact of daily changes in seasonal AOD on DARE in different KG climate zones.

early morning and evening. From an interannual perspective, as shown in Fig. 9, the differences are smallest between 8:00 a.m. and 4:00 p.m., especially during the period from 10:00 a.m. to 2:00 p.m., where the results are particularly favorable.

Based on the Köppen–Geiger (KG) climate zone system, global sites have been categorized, and analyses have been conducted for four selected regions: tropical climates (A), arid climates (B), temperate climates (C), and continental climates (D), as shown in Fig. 8(b). In Region Tropical Climates, the AOD values at 8:00 a.m. and from 11:00 a.m. to 1:00 p.m. are well-suited for representing the daily average AOD in clear-sky radiation effect calculations. For Region Arid Climates, the AOD values at 8:00 a.m., 9:00 a.m., 1:00 p.m., and 4:00 p.m. are particularly representative. Significant AOD values are observed in Region Temperate Climates at 7:00 a.m., 11:00 a.m., 12:00 p.m., and 4:00 p.m.. In Region Continental Climates, the AOD values from 9:00 a.m. to 4:00 p.m. are consistent and appropriate for representing the daily average AOD in clear-sky radiation effect calculations, with a slight decrease noted at 1:00 p.m.

This study focused on comparing the MODIS, MISR, and VIIRS AOD products with overpass times at 10:30 and 13:30 local time to analyze their representativeness of daily mean values. For this assessment, satellite-derived AOD (averaged within ± 1 hour of the overpass times at 10:00 and 13:00) was compared with DARE calculated from AERONET daily mean AOD, as illustrated in Fig. 10. The results showed good agreement for both overpass times, with 10:30 ($R = 0.96$, RMSE $= 5.31$) and 13:30 ($R = 0.95$, RMSE $= 5.38$).

Fig. 11 presents the differences in clear-sky DARE resulting from using hourly AOD averages as substitutes for daily AOD averages across various aerosol types. In dust-dominated regions, the clear-sky DARE calculated using hourly AOD averages ranged between -30 and -25 W·m^{-2}, with deviations from the DARE derived from AERONET daily AOD averages remaining below 5.46 W·m^{-2}. Morning hours exhibited larger biases compared to afternoon periods.

In biomass burning-dominated regions, the clear-sky DARE calculated from hourly AOD averages spanned -40 to -25 W·m^{-2}, showing greater deviations from daily-averaged DARE values. The maximum discrepancy reached 10.61 W·m^{-2} during evening hours, though differences remained relatively small between 7:00 and 17:00 local time. For anthropogenic aerosol-dominated regions, clear-sky DARE values stabilized around -30 W·m^{-2} with minimal fluctuations. Evening DARE biases slightly exceeded morning deviations but stayed below 5.72 W·m^{-2} overall. Between 8:00 and 16:00, DARE deviations were notably smaller, particularly during midday when variability reached its most stable level.

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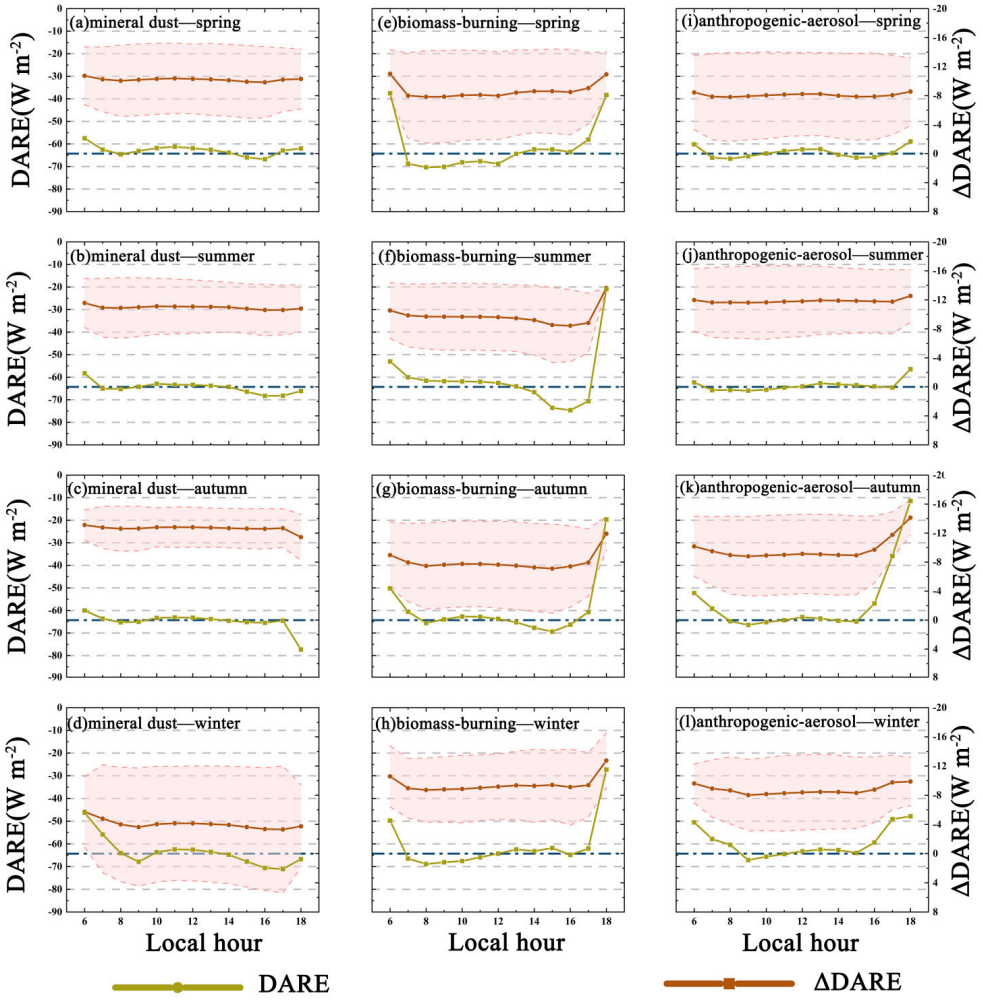


Fig. 14. Differences in DARE calculated based on the seasonal daily variations of AOD for different aerosol types.

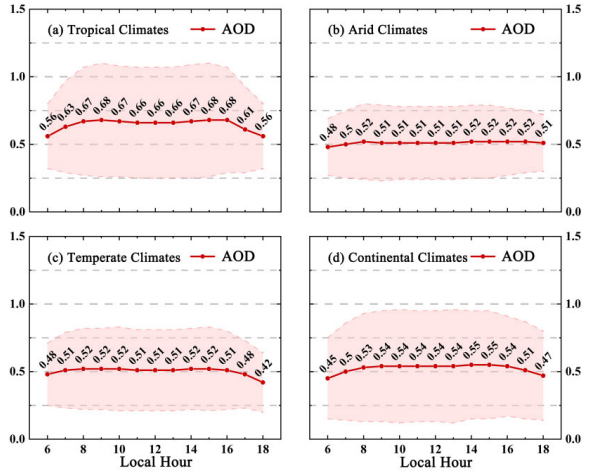


Fig. 15. Daily changes in AOD in different KG climate zones.

D. Impact of Seasonal Daily Variations in AOD on Daily DARE

Aerosols behave differently in different seasons, usually with smooth daily changes in the fall and larger changes in the spring and summer. As shown in Fig. 12, the DARE for the whole day using the hourly AOD alternative daily averages for different seasons shows a certain regularity in different time periods. In spring, the DARE fluctuates mainly in the range of −30 to −40. The closest results were obtained using hourly AOD between 10 a.m. and 2 p.m. compared to the radiative effect calculated using daily average AOD. The DARE during the summer months, on the other hand, fluctuated in the range of −30 to −20. The results obtained using hourly AOD during the period 7 a.m. to 2 p.m. were the closest match compared to the radiative effects calculated using daily AOD. The morning AOD performed better in the summer compared to the spring. The DARE was mainly stable around −30 in the fall. Differences in DARE were relatively large in the fall when using hourly AOD instead of daily average AOD to calculate DARE compared to the other seasons. DARE was lower in winter, hovering mainly around −40. In contrast, the results obtained using the hourly AOD were closest to those calculated using the daily average AOD during the period from 11 a.m. to 4 p.m., which may be related to the calmer weather variations in winter.

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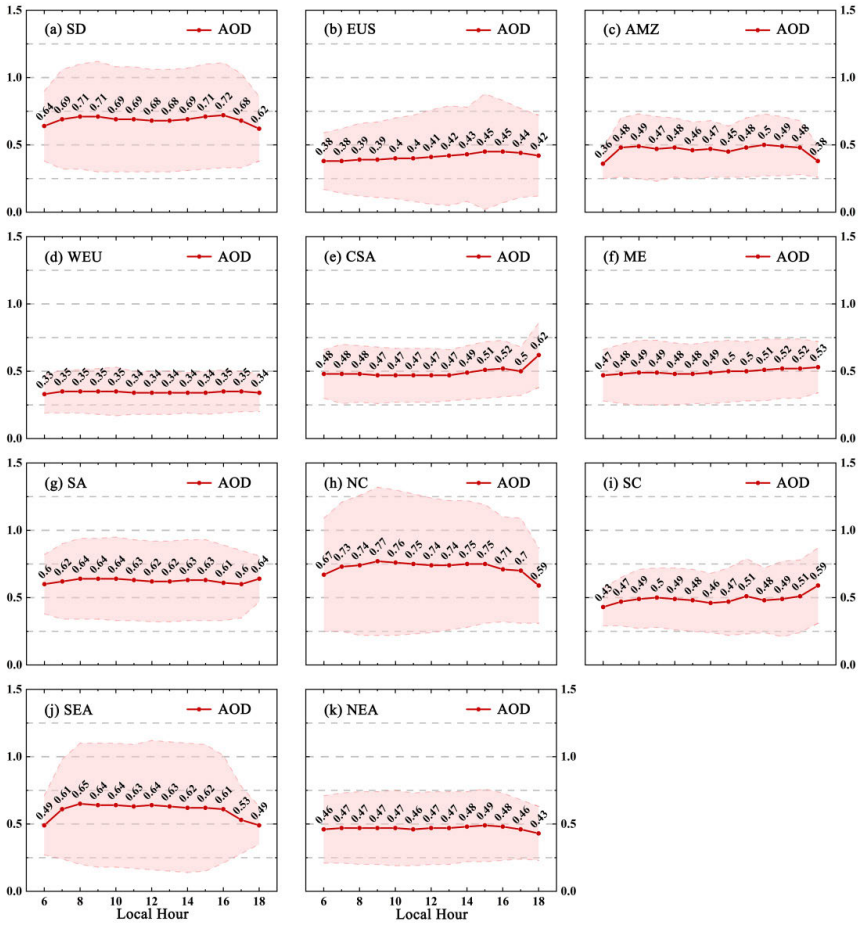


Fig. 16. Daily changes in AOD in 12 different geographical divisions.

The daily variation of aerosols varies under different climatic influences. Arid climatic regions are prone to localized thermal circulation due to higher daytime temperatures and uneven heating of the ground, resulting in more pronounced daily changes in aerosols. In contrast, temperate oceanic climatic regions are mainly regulated by the ocean, the weather is more stable, and aerosol changes are gentle. Continental climates have high summer temperatures and strong convective movements, which make aerosols easy to diffuse and dilute, while in the winter when temperatures are low and influenced by human activities, daily aerosol variations are usually more significant.

Fig. 13 shows the effect of seasonal and daily variations on DARE in different KG climate zones. In the tropical climatic zones, the effect of daily variation of AOD on DARE is more pronounced in the four seasons, ranging from -8 to 20 W·m^{-2}. The fluctuations are larger in the morning and evening. Temperate, arid, and continental climates had very flat day-to-day changes in the spring and summer, ranging from -2 to 4 W·m^{-2}. Stronger fluctuations are seen in the morning and evening in the fall. Winter fluctuations are relatively small in temperate regions. In the winter in continental climates, the use of 6-point AOD instead of one day's AOD caused the largest difference of 20 W·m^{-2}.

Fig. 14 illustrates the seasonal diurnal variations of clear-sky DARE differences calculated based on AOD seasonal patterns in the dust-, biomass burning-, and anthropogenic aerosol-dominated regions from 2015 to 2019. The results reveal distinct seasonal diurnal characteristics in DARE across different aerosol types.

In dust-dominated regions, clear-sky DARE exhibited significant seasonal variations. Spring and summer showed the most stable DARE values, remaining around -30 W·m^{-2}, with minimal diurnal fluctuations in summer. Autumn displayed lower DARE values compared to spring and summer, featuring an asymmetric distribution pattern with higher values in the morning and lower values in the evening. Winter demonstrated the most pronounced DARE fluctuations, averaging around -50 W·m^{-2}, with relative stability only between 08:00 and 14:00. Compared to DARE calculated from daily AOD averages, winter exhibited smaller errors between 10:00 and 14:00, while other seasons showed reduced errors during 07:00–15:00.

In biomass burning-dominated regions, clear-sky DARE remained stable between -30 and -40 W·m^{-2} in spring and summer, slightly decreased in autumn, and moderately increased in winter. Diurnal DARE errors relative to daily AOD-derived values were most pronounced during early morning and evening hours. Summer evenings recorded the

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largest error (13.47 $W{\cdot}m^{-2}$), while autumn mornings showed slightly higher deviations than summer. Winter exhibited minimal errors between 11:00 and 15:00, with evening errors lower than those in summer and autumn.

For anthropogenic aerosol-dominated regions, DARE remained most stable in spring and summer (-30 ± 3 $W{\cdot}m^{-2}$), with negligible diurnal variations. Post-15:00 DARE values increased significantly in autumn, accompanied by pronounced error growth during 15:00–18:00. Winter DARE fluctuated between -40 and -30 $W{\cdot}m^{-2}$, with elevated errors in morning and afternoon periods, while errors remained smaller between 10:00 and 15:00.

Fig. 15 displays the mean diurnal variations of AOD across different climate zones. The results show that arid climate zones exhibit the smallest daily AOD fluctuations (0.48–0.52). Tropical climate zones maintain higher AOD levels throughout the day with larger variations (0.56–0.68). Temperate and continental climate zones also demonstrate relatively stable AOD patterns, ranging between 0.42–0.52 and 0.45–0.55, respectively.

Fig. 16 reveals significant regional heterogeneity in the diurnal variations of AOD across global representative regions. The study identified notable regional disparities, with high-variability regions such as NC showing the largest AOD fluctuations (0.59–0.77). In NC, AOD surged sharply by 15% (from 0.67 to 0.77) during early morning hours (06:00–08:00), primarily driven by anthropogenic activities. This was followed by AMZ, SEA, and SC, which exhibited ΔAOD values of 0.14, 0.16, and 0.16, respectively. In contrast, low-variability regions included WEU, where AOD remained stable throughout the day ($\Delta\text{AOD} = 0.02$, range: 0.62–0.64), reflecting a dynamic balance between urban emissions and dispersion. SA showed minimal fluctuations (0.60–0.64), while the Middle East (ME) and northeastern Asia (NEA) displayed very small ΔAOD values (0.06 and 0.04), indicating that dust events and high-latitude radiative changes in these regions did not trigger significant AOD variations. SD exhibited a rapid morning AOD rise, maintaining consistently high levels throughout the day (0.69 ± 0.03), highlighting the persistent influence of daytime dust activity.

IV. Conclusion

The temporal representativeness of satellite aerosol products has extremely important implications in the study of the relationship between aerosols and atmospheric radiation. The present study analyses the impact of satellite aerosol products in representing mean aerosol values over time by comparing the effects of clear-sky radiation modeled using satellite aerosol products and AERONET site AOD data.

MISR shows the best performance in representing instantaneous matches, with an R value of 0.8833 and an RMSE of 19.4115 $W{\cdot}m^{-2}$. MODIS follows with R values of 0.8033 and 0.7741, and RMSE of 12.5886 and 13.3884 $W{\cdot}m^{-2}$, respectively. The VIIRS satellite performs next best, achieving an R value of 0.8013 and an RMSE of 16.4216 $W{\cdot}m^{-2}$. When using satellite AOD data instead of daily average AOD values for radiative effect simulations, MISR shows significant improvement, with an R value of 0.8593 and an RMSE of 13.3719 $W{\cdot}m^{-2}$. MODIS also performs well, achieving R values of 0.394 and 0.8374, and RMSE values of 11.1417 and 10.9722 $W{\cdot}m^{-2}$, indicating strong correlations and relatively low errors. VIIRS, though less effective, still demonstrates a respectable R value of 0.7885 and an RMSE of 14.3992 $W{\cdot}m^{-2}$. Globally, the simulation performs best in WEU. In contrast, in the SD and Central South America (CSA), all four satellites—MISR, MODIS Aqua, MODIS Terra, and VIIRS—tend to underpredict AOD values, suggesting a systematic bias in these regions.

We also assess the representativeness of using hourly AOD values instead of daily average AOD values for calculating clear-sky radiative effects. The results show that DARE values from 8:00 a.m. to 2:00 p.m. exhibit the least variation (less than 0.6 $W{\cdot}m^{-2}$), while the largest differences occur in the early morning and evening.

This study undertakes a comprehensive global-scale exploration into the disparities between utilizing satellite AOD products as substitutes for time-averaged values in radiative effect simulations, which would provide support for future satellite transit time design. However, owing to constraints imposed by both temporal and spatial factors, the number of sites fulfilling the requisite criteria for radiative simulation remains limited. It is noteworthy that differences in spatial resolution among various satellite products can also introduce errors. Aerosol distribution exhibits pronounced spatiotemporal heterogeneity, and due to resolution limitations, satellite-retrieved AOD products inherently contain spatial uncertainties. Schutgens et al. [49] demonstrated that spatial representativeness errors in satellite products may remain significant even after extensive data averaging, with these errors exceeding typical measurement uncertainties. Chen et al. [50] systematically analyzed the spatial representativeness of AERONET ground-based observations compared with satellite measurements from Terra/Aqua MODIS and Himawari-8 AHI at six typical sites across Asia. In future research, the impact of spatial representativeness errors in satellite AOD products on the ADARE warrants further investigation.

Besides AOD, other critical parameters such as SSA can also exert substantial impacts on the DARE. Previous studies have demonstrated that the diurnal variation of solar zenith angle (SZA) significantly influences atmospheric radiative transfer processes [52]. As SZA increases, the extended transmission path of solar radiation through the atmosphere enhances aerosol-radiation interactions, thereby substantially affecting radiative balance. Chen et al. [53] observed that with increasing SZA, the cooling effect of all aerosol types at the top of atmosphere (TOA) initially intensifies before weakening. Xia et al. [54] through analyzing global variations in radiative forcing efficiency (RFE) of major aerosol types with SZA, reported that the maximum TOA RFE cooling effect occurs when SZA ranges between 0.3 and 0.4. Logothetis et al. [55] study identifies SSA as the second most influential factor governing ADRE, following AOD. Research by Lu et al. [56] reveals a pronounced SSA dependence in surface-layer DARE across diverse aerosol types. Particularly for absorbing aerosols, their direct radiative forcing displays

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greater sensitivity to SZA and surface albedo fluctuations compared to other types. Lu et al. [56] found that minor variations in SSA can induce significant fluctuations in DARE, while changes in ASY exhibit limited impact on total atmospheric heating by aerosols. Specifically, when ASY varies within $\pm 30\%$, the corresponding DARE variation remains confined to -5% to $+2\%$. Further investigation is required to quantify how uncertainties in additional key optical parameters (including but not limited to SSA and asymmetry factor) influence DARE estimates.

ACKNOWLEDGMENT

The authors are grateful to the MODIS, VIIRS, MISR, and Aerosol Robotic Network (AERONET) Teams for their efforts to make the data available.

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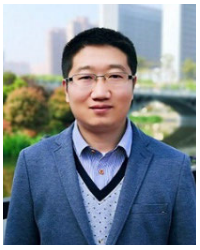
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